



RMKT EMAIL AUTOMATION · INTERNAL TOOL

EmailAuto Studio

Brief in → on-brand, email-safe HTML out. AI-assisted email production for BraGoddess, GentsLux, LuxFitting & SantaFare — generate, preview, and push straight into SendGrid as a Design or Dynamic Template.

BraGoddess

GentsLux

LuxFitting

SantaFare

OVERVIEW

What it does

THE PROBLEM

Hand-built emails, slowly

The team manually wrote copy, hand-built HTML per brand/tier/segment, then ran a cleanup script before loading templates into SendGrid — repetitive and error-prone.

THE SOLUTION

One guided studio

A campaign brief produces every tier × segment variant with proven, on-brand copy and SendGrid-ready HTML — previewed, QA'd, saved, and synced in a few clicks.

BUILT ON PROOF

The winning playbook

Prompts encode the analysis of 46 real campaigns: brand personas, the Hook Contract, and win/fail guardrails — so output follows what actually converts.

CLOSES THE LOOP

Straight into SendGrid

Push as a Design, or run the team's cleanup pass and create a Dynamic Template — returning the `d-...` Template ID for sends.

HOW IT WORKS

The studio pipeline



Playbook-driven, on-brand copy

- ✓ Brand **personas** — Sandra · Jordan · Adele · Mary
first-person voice per brand
- ✓ **Hook Contract** — one locked promise
carried through subject, body, CTA, preheader
- ✓ **Win/fail guardrails**
named micro-story openers, no banned phrases, 2–4 word CTAs
- ✓ Brand **subject / urgency / preheader** formulas
e.g. SantaFare suspended-loop, GentsLux curiosity-gap
- ✓ Corrected **segment IDs** per brand
tier × segment variant matrix, keyed A21, B62...
- ✓ **Prompt transparency**
see & edit every prompt before it's sent; what you see is sent

SendGrid sync · Accounts · History · Admin

SENDGRID V3

Design & Dynamic Template

Module-format HTML that round-trips into SendGrid's editor. Dynamic Template runs the team's clean/optimize/QA pass (the Apps Script, ported) and returns the `d-...` ID.

IMAGE WORKFLOW

Paste SendGrid CDN URLs

Per-block image panel (logo optional). No upload backend needed — exactly the team's existing workflow.

ACCOUNTS & HISTORY

Save & reopen campaigns

Each user saves whole-campaign generations and reopens them later. Row-Level Security scopes data per user.

SUPER ADMIN

Approval-gated access

New signups are pending until an admin approves. Admins activate/deactivate users and reset passwords.

TECH STACK

How it's built

Framework	Next.js 15 (App Router, TypeScript) · React 19 · Tailwind CSS v4
AI	Anthropic Claude (claude-sonnet-4-6) via the Anthropic SDK · prompt-cached system prompts · one call per tier
Email API	SendGrid v3 (@sendgrid/client) — Designs + Dynamic Templates
Auth + Database	Supabase (Postgres + Auth) · email/password · Row-Level Security
Hosting	Vercel — auto-deploy from <code>main</code> ; secrets in encrypted env vars
Repository	<code>minhhvle-crossian/emailauto-studio</code> (private)

SECURITY

Verified on production

✓ Authenticated paid routes

generation + SendGrid sync require an active user (401 otherwise)

✓ Admin authorization

token + is_admin + active; non-admin → 403

✓ Row-Level Security

users see only their own data; no self-grant of admin

✓ No secret leakage

service-role / SendGrid / Anthropic keys are server-only

✓ Sandboxed preview

email rendered in a no-script iframe; inputs escaped

✓ HTTPS + HSTS & security headers

X-Frame-Options, nosniff, Referrer/Permissions-Policy

Recommended next: rotate shared keys · rate-limit generation · optional CSP & signup domain allowlist.

GOVERNANCE

AI assists — a human always decides

The studio speeds up production, but a person stays in control at every gate. Nothing reaches a recipient without human review and a human hitting Send.

GATE 1 · PROMPTS

See the prompts

Every prompt is shown — and editable — before it's sent to the model.

GATE 2 · COPY

Edit the copy

AI drafts; the marketer reviews and edits every variant before it's used.

GATE 3 · PREVIEW

Pass pre-flight

Win-pattern QA must clear; the human fixes any flags before export.

GATE 4 · SENDGRID

Human sends

The app only creates drafts/designs — a person clicks Send in SendGrid.

No autonomous sending. Access is approval-gated, and the final send is always a deliberate human action inside SendGrid.

FEE STRUCTURE

Cost to run

PER CLAUDE API CALL (ONE TIER)

~\$0.015–0.02

Output tokens dominate the cost; the brand system prompt is cached, so repeat tiers are cheaper.

PER FULL CAMPAIGN

~\$0.03–0.10

A few tiers × segments. Example: 2 tiers × 2 segments = 2 calls ≈ **\$0.033**.

CLAUDE (VARIABLE)

~\$2–5 / mo

at ≈ 50 campaigns / month

HOSTING

~\$0–45 / mo

Vercel (Hobby free / Pro \$20) · Supabase (Free / Pro \$25)

SENDGRID

Existing

uses your current plan — no added cost

Estimated at Claude Sonnet rates (\$3 / \$15 per M tokens; cache read \$0.30/M). Marginal cost ≈ a few cents per campaign.

Where it is — and where it can go

GAPS TODAY

- △ Sending & scheduling are manual (human sends in SendGrid)
- △ No performance / analytics loop yet
- △ Images pasted manually — no upload or asset library
- △ No rate-limiting / spend caps
- △ No A/B variant tracking

POTENTIAL

- ↑ End-to-end: trigger → generate → schedule → send
- ↑ Closed-loop optimization from real SendGrid stats
- ↑ Self-serve image upload + shared brand asset library
- ↑ Scale to more brands, markets & lifecycle flows
- ↑ Auto-A/B that feeds winning patterns back into the playbook

IMPROVEMENT

Near-term improvements

- ✓ Rate-limit generation + spend caps
cap Claude cost per user
- ✓ Rotate & vault API keys; audit logging
security hygiene
- ✓ Analytics agent
pull SendGrid stats → weekly insight reports
- ✓ SendGrid Single Send draft + scheduling
closer to one-click send (human approves)
- ✓ A/B subject & variant testing
measure, then bake wins into the playbook
- ✓ Brand image libraries + richer history
search, duplicate & reuse assets

STATUS

Shipped & live

LIVE

Production

emailauto-studio.vercel.app — auth, studio, history, admin all working.

BACKENDS

Connected

Supabase (auth + RLS) and SendGrid (Design + Dynamic Template) verified end-to-end.

SECURITY

Hardened

Re-checked on the live deployment; headers and route auth in place.

From hand-built emails to a guided, governed, one-click studio — in production.